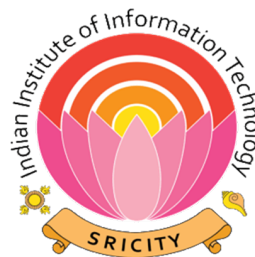


Roll Number:

Name:

Section:



Indian Institute of Information Technology, Sri City, Chittoor
Spring Semester Schedule Quiz 2, April-2026
UG-1, Second Sem

Name of the Exam: P & S (S2)

Duration: 20 Minutes

Apr 15, 2026

Max. Marks: 10

Instructions:

1. Calculators are allowed.
 2. All questions are mandatory and carry equal mark.
 3. Tick only the right option. If we find two ticks for one question, '0' marks will be awarded.
 4. Rough work should not be done in the question paper.
 5. Return the question paper along with the rough sheet before leaving the exam hall.
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1. While tossing 100 coins, let H be the number of heads. What is the probability that $40 < H < 60$ using Chebyshev's inequality?

- (a) 0.50
- (b) 0.75
- (c) 0.33
- (d) 0.67

Answer:(b)

2. Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with mean 5 and variance 9. As $n \rightarrow \infty$, $\frac{X_1^2 + X_2^2 + \dots + X_n^2}{n} \xrightarrow{p} c$ for some constant c . Then the value of c is:

- (a) 25
- (b) 34
- (c) 14
- (d) 28

Answer:(b)

3. If $E[X|Y] = 2Y + 1$, then $E[X]$ is

- (a) $2E[Y]$
- (b) Cannot be determined

(P.T.O.)

(c) $2E[Y] + 1$

(d) $E[Y] + 2$

Answer: (c)

4. Let $X_1, \dots, X_7 \sim N(0, \sigma^2)$ and $Y_1, \dots, Y_9 \sim N(0, \sigma^2)$ be two independent random samples. Define

$$U = \sum_{i=1}^7 X_i^2, \quad V = \sum_{j=1}^9 Y_j^2,$$

and consider the statistic

$$F = \frac{(U/7)}{(V/9)}.$$

Further, define the transformed variable

$$T = \frac{1}{F}.$$

Then T follows an F -distribution with degrees of freedom $(---, ---)$. (Enter your answer as an ordered pair)

Answer: (9, 7)

5. Let X and Y be independent random variables, each following the uniform distribution on $(0, 1)$. Which of the following best describes the probability density function of $Z = X + Y$?

(a) Exponential distribution

(b) Triangular distribution

(c) Uniform distribution

(d) Normal distribution

Answer: (b)

6. ****Let $f_{X,Y}(x, y) = xy$, $0 < x < 2$, $0 < y < 2$. Which of the following is correct?**

(a) $E[XY] = E[X] + E[Y]$

(b) X and Y are independent

(c) $\text{Cov}(X, Y) = 0$

(d) X and Y are not independent

Answer: ()**

7. Let $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$. Consider the sample mean $\bar{X}$ and sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

and the maximum likelihood estimator (MLE) of variance

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Which of the following statements are correct?

- (a) $\bar{X}$ is an unbiased estimator of μ , and S^2 is an unbiased estimator of σ^2 .
- (b) $\bar{X}$ is a consistent estimator of μ .
- (c) $\hat{\sigma}_{\text{MLE}}^2$ is biased but consistent for σ^2 .

- (a) Only (a) is correct
- (b) (a), (b), and (c) are all correct
- (c) (a) and (b) are correct
- (d) (b) and (c) are correct

Answer:(b)

8. The average age of the cricket players is 28.3 years and has standard deviation of 2.3 years. If we can assume that ages are normally distributed, what is the probability that the average age of 10 randomly selected players is less than 27 years?

- (a) $\Phi(-0.57)$
- (b) $\Phi(-1.79)$
- (c) $1 - \Phi(-1.79)$
- (d) $\Phi(0.57)$

Answer:(b)

9. Which of the following statements are correct?

- (a) The estimator $T_n = \bar{X}_n + \frac{1}{n}$ is biased and consistent.
- (b) The estimator $T_n = X_1$ is unbiased but not consistent.

- (a) Both (a) and (b) are correct
- (b) Neither (a) nor (b) is correct
- (c) Only (a) is correct
- (d) Only (b) is correct

Answer:(a)

10. Let $U \sim N(0,1)$ and $V \sim \chi_\nu^2$ be independent random variables. Define

$$T = \frac{U}{\sqrt{V/\nu}}.$$

Let $\nu = 8$. Consider the transformed variable

$$W = \frac{T^2}{T^2 + \nu}.$$

Then W follows a Beta distribution $\text{Beta}(a, b)$. Find the values of (a, b) .
(Enter your answer as an ordered pair (---, ---))

Answer: $(\frac{1}{2}, 4)$
